

Homework 2: Math through September 30

Name _____

IT 520 · Fall 2026

Purpose: practice all calculation methods used in class so far. Show the setup, substituted values, units, and arithmetic for every part. A final answer without readable supporting work receives no credit.

- Submit one PDF containing this sheet or equivalent clearly labeled work.
- Handwritten or typed work is acceptable. Scan or photograph handwritten pages clearly and combine them into one PDF. Scanning is available at Kingsbury Library.
- Estimate: 75–90 minutes. Numerical scenarios are hypothetical.

1. A sensor hub's processor

A sensor hub runs at **2.4 GHz**. Processing one batch requires **9.6 billion cycles**.

$$\text{cycle time} = 1 \div \text{clock rate} \quad \cdot \quad \text{execution time} = \text{cycles} \div \text{clock rate}$$

$$1 \text{ GHz} = 1 \text{ billion cycles/second}; 1 \text{ nanosecond} = 10^{-9} \text{ second}$$

a. Find one cycle's time in seconds and nanoseconds. **b.** Find the batch's processor execution time. **c.** In one sentence, explain why clock rate alone cannot predict the whole batch's elapsed time.

2. A backup pipeline

One backup takes 60 minutes: scan files, 12 min; compress, 18 min; upload, 30 min. A new processor makes *compression* three times as fast. The other stages do not change.

a. Find the new compression time and new whole-job time. **b.** Find the compression speedup and whole-job speedup. **c.** If compression time approached zero, find the limiting whole-job time and the largest possible whole-job speedup from changing only compression.

Systems: rates, time, and limits

3. The help desk window

During a 30-minute window, one help-desk agent completes **18 tickets** and is busy for **24 minutes**. An average of **1.8 tickets** are inside the help-desk boundary (waiting or being served).

$$X = C/T \quad \cdot \quad S = B/C \quad \cdot \quad U = X \cdot S \quad \cdot \quad N = X \cdot R$$

a. Find throughput X in tickets/minute and tickets/hour. **b.** Find service time S . **c.** Find utilization U two ways: B/T and $X \cdot S$. **d.** Use Little's Law to find average response time R . **e.** Find average waiting time, $R - S$.

Interpretation: Which is longer for an average ticket—working time or waiting time? Support the answer with your results.

4. Find the web service bottleneck

In a 60-second observation, a web service completes 15 requests. Its one CPU is busy for 30 seconds, one disk for 45 seconds, and one network link for 15 seconds.

Resource	Busy time B	Service demand $D = B/C$
CPU	30 s	
Disk	45 s	
Network	15 s	

a. Find each service demand in seconds/request. **b.** Name the bottleneck and calculate its throughput ceiling, $1/D_{\max}$, in requests/second and requests/minute. **c.** A proposed CPU upgrade halves CPU demand. Recompute the tallest demand and ceiling. Should this be the first upgrade if the goal is a higher ceiling? Explain.

Integers: width, notation, and signed meaning

5. Choose a counter width

A lab counter must represent every count from **0 through 90 inclusive**.

a. How many distinct values are required? **b.** Find the minimum width n . State the number of patterns, 2^n , and the largest unsigned value. **c.** How many patterns remain unused? **d.** Decode the seven-bit report 1011010_2 as unsigned decimal. Is it a valid count for this counter?

6. Read a packed color byte

A device sends one byte as $5C_{16}$.

a. Convert it to binary as two space-separated nibbles. **b.** Convert it to decimal by showing the binary place values that are on. **c.** A second device sends $1010\ 0111_2$. Convert that byte to hexadecimal.

7. A five-bit signed temperature adjustment

The field uses **five-bit two's complement**. A received pattern is 11101_2 .

a. State the signed range. **b.** Read 11101_2 as unsigned, then as two's complement. Show the rule you used. **c.** The current adjustment is $+7$ (00111_2) and software adds $+11$ (01011_2). Add the patterns, keeping five bits. Does the true result fit? Name what happened and explain the stored pattern's signed reading.

Binary conversions past the point

8. Assign an archive code

An archive assigns decimal code **53**. Convert 53 to binary using division by 2. Fill the quotient and remainder at every step, then read the remainders last to first. Check your result by binary place value.

Divide	Quotient	Remainder
53 ÷ 2		

53 = _____₂ Check: _____

9. Measurements between whole numbers

- a.** A timer reports 10.101_2 seconds. Convert it to decimal by naming each nonzero place value.

- b.** Convert decimal **0.6875** to binary by repeated doubling. Stop when the kept fraction is zero, and check by place value.

c. Try to convert decimal **0.4** by repeated doubling. Stop when a kept fraction repeats. Record enough rows to show the loop.

Double	Whole-number bit	Keep the fraction
$0.4 \times 2 =$		

Write its binary form using an ellipsis: _____. Which kept fraction repeats? _____

- d.** A computer has a fixed number of bits. Explain why it stores a nearby value rather than exactly 0.4, and predict whether adding or comparing decimal fractions can expose that approximation.

Submission check: one PDF; all nine problems labeled; formulas or tables visible; units on measurements; explanations answer the question asked.